SREENAND O

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PROFESSIONAL SUMMARY

Performance-driven AI & Machine Learning Engineer with hands-on expertise in developing autonomous multi-agent workflows, hybrid retrieval-augmented generation (RAG) pipelines, and edge computer vision architectures. Proficient in Python, PyTorch, LangChain, FastAPI, and Next.js. Skilled in fine-tuning foundation models, optimizing vector retrieval accuracy, and transitioning machine learning research into production-grade systems.

TECHNICAL SKILLS

• Core AI & ML: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Scikit-Learn, CNNs, Transformers, Model Quantization, TensorRT, CUDA

• Generative AI & Agents: LangChain, LlamaIndex, ReAct Loop Architecture, Multi-Agent Orchestration, Semantic Routing, Prompt Engineering

• Vector Databases: Qdrant, ChromaDB, Pinecone, Hybrid Search (Dense + BM25), Cross-Encoder Contextual Reranking

• Languages & Backend: Python, TypeScript, JavaScript, SQL, C++, FastAPI, Next.js, Docker, RESTful APIs, Git, Linux

EXPERIENCE & PROGRAMS

Prime Minister Internship Scheme | AI / Systems Intern

Government of India Initiative | 2025 – Present

• Researched automated data ingestion workflows and machine learning-driven validation pipelines to streamline administrative data extraction.

• Implemented structured Python automation routines to inspect datasets, validate schemas, and monitor pipeline error rates.

• Documented system operational metrics and contributed to architectural feasibility reports for digital service scaling.

K-DISC Learning Program | Machine Learning Fellow

Kerala Development and Innovation Strategic Council | 2024 – 2025

• Selected for an intensive fellowship covering applied computational intelligence, predictive modeling, and edge inference deployment.

• Designed modular deep learning pipelines for computer vision and text classification tasks, optimizing model convergence and F1-score.

• Collaborated with peer engineering cohorts to build prototype solutions addressing regional technical and logistical challenges.

FEATURED PROJECTS

Autonomous Multi-Agent Research System (LangChain, FastAPI, ChromaDB, Python)

• Engineered a multi-agent architecture utilizing the ReAct framework to divide research queries into automated subtasks.

• Built schema enforcement guardrails with Pydantic, achieving a 96.4% factual synthesis benchmark on domain-specific question sets.

• Constructed an asynchronous FastAPI streaming backend delivering step-by-step reasoning outputs via Server-Sent Events (SSE).

Enterprise Hybrid Semantic RAG Engine (Qdrant, Hugging Face, FastAPI, Docker)

• Implemented an industrial retrieval-augmented generation system combining dense vector embeddings with BM25 sparse keyword ranking.

• Integrated a cross-encoder contextual reranker (bge-reranker-large), lifting top-5 retrieval precision and recall to 98.1%.

• Containerized the retrieval service using Docker to standardize environment configuration and reduce deployment cold starts by 35%.

Real-Time Computer Vision & Edge Gesture Engine (PyTorch, YOLOv8, OpenCV)

• Designed a low-latency edge vision pipeline tracking spatial coordinates and motion gestures at 60+ FPS with sub-10ms processing latency.

• Applied post-training quantization (FP16) to optimize neural network inference speed across consumer hardware.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

APJ Abdul Kalam Technological University, Kerala | Expected 2027

• Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning Foundations, Linear Algebra, Database Management Systems.

CERTIFICATIONS & ACTIVITIES

• IEDC Summit: Active Delegate & Contributor in applied artificial intelligence and technical venture workshops.

• Open Source Contributor: Actively developing and publishing AI repositories, microservice templates, and agentic workflows.